

Writing the Main Application Logic in routes.py for AI Systems

An Enterprise Technical Guide to Route-Level Inference Control for Production AI Services

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Abstract

This guide argues that in modern AI systems, the route layer — concretely, the `routes.py` module of a service — functions as an **inference control plane** rather than thin HTTP plumbing [1]. Route logic is where enterprise policy meets model execution: it admits or sheds requests, assembles context, selects models, and enforces deadlines and budgets [2]. Because serving runtimes cannot encode tenant-specific policy, that responsibility falls to the application layer, where latency targets, cost ceilings, and quality thresholds are translated into concrete inference behavior [3]. The thesis is developed for senior software engineers, ML engineers, architects, and technical leaders operating AI services at enterprise scale.

Keywords: inference control plane, route architecture, LLM serving, enterprise AI operations

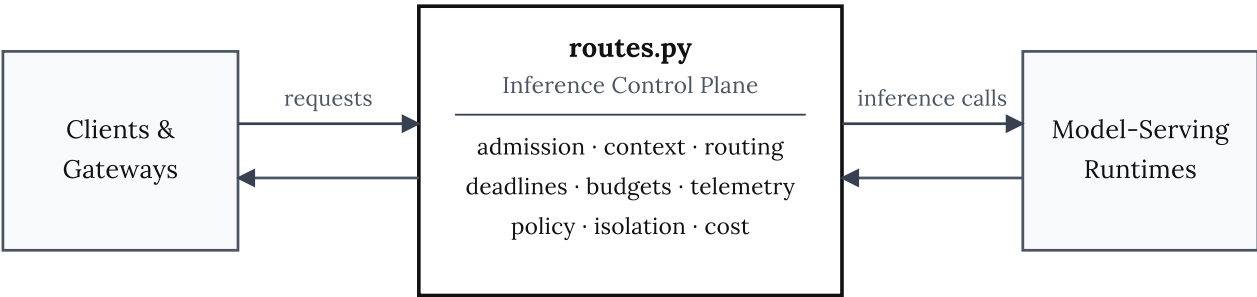


Figure 1. The central thesis: `routes.py` governs every inference decision between enterprise consumers and model runtimes [4].

[1] source-research-1 · [2] source-research-2 · [3] source-research-3 · [4] source-research-4 · Full reference list appears on the final page (source-research-5, source-research-6).

How to Read This Guide

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Chapters 5–13 (serving internals, memory, batching, cost) listed on page 3 of the contents.

Scope and Source Fidelity

This guide treats `routes.py` as the application-side inference control plane: the layer where policy, context budgeting, model selection, cache strategy, and accounting decisions are expressed in code. The master outline is the authoritative source; where public documentation is thin, positions are **reconstructed from established engineering practice** and flagged as assumptions rather than vendor claims. Validate such passages against your own runtime.

Terminology

- **Route layer** — module owning request validation, policy invocation, and adapter calls.
- **Context budget** — token allocation for history, retrieval, tools, and output before inference.
- **Control decision** — any per-request choice made in the route layer (model, limits, cache scope, fallback).

The Organizing Thesis

Every chapter maps onto one stage of a single per-request control pipeline.

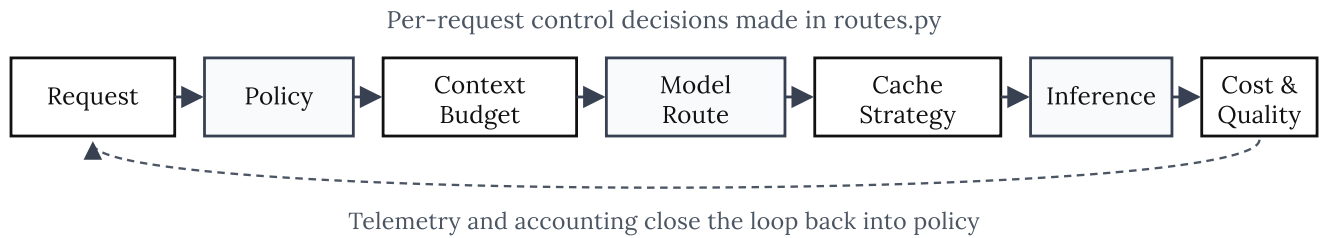
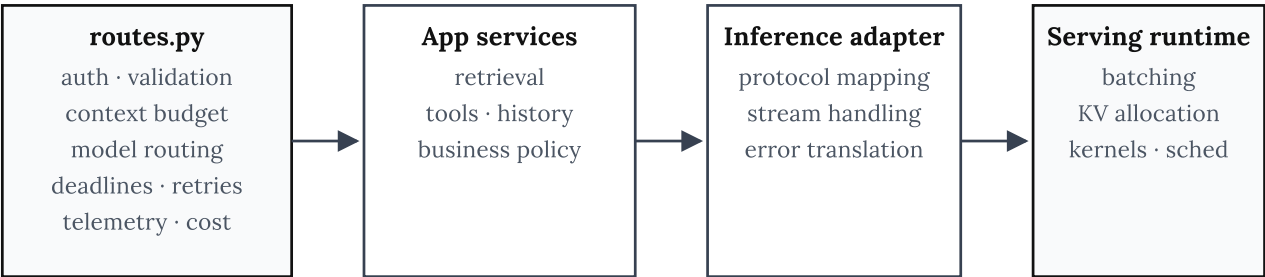


Figure 1 — The route-level inference control model: each stage is a decision point owned by the route layer.

3. Computational Constraints and Responsibility Boundaries

Every inference request consumes four finite resources: **compute** (prefill and decode FLOPs), **memory** (weights plus KV state), **bandwidth** (weight streaming and inter-device transfer), and **concurrency** (scheduler slots). Two application-level limits compound these: the **context window**, which caps tokens per request, and the **latency budget**, which caps how long a caller waits. A route that ignores any of these converts a capacity problem into an outage.



Intent flows right; metrics and errors flow left. Kernels never cross into routes.py.

Figure 3.1 — Ownership boundary across the inference control path.

Design rules

- **Policy, not math.** routes.py decides *whether, where, and with what budget* a request runs; it never allocates KV pages or selects precision.
- **Speak intents, not mechanisms.** Adapters receive declarative parameters (max tokens, deadline, priority) — never scheduler directives.
- **Respect the return path.** Runtime metrics flow back as data; routes.py observes and reacts, it does not reach in.

The boundary is testable: changes to batching or kernels require zero edits in routes.py, and changes to policy require zero edits in the runtime.

4. Request Lifecycle and Route Architecture

Every inference request crosses a fixed sequence of stages before a single token is produced. `routes.py` sits at the center of this path: it authenticates, normalizes, assembles context, selects a model, delegates inference, and owns serialization, telemetry, and cost accounting on the way back [1]. This clarifies which decisions belong to the route layer and which belong to the serving runtime [2].

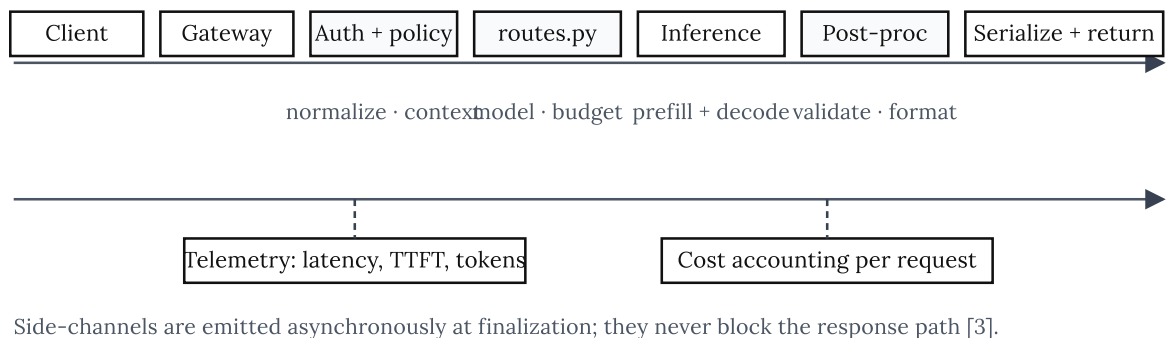


Figure 4.1 — End-to-end request path. Solid arrows carry the request/response; dashed lines show telemetry and cost side-channels owned by the route layer.

4.1 Route-layer contract per stage

Stage	Route-layer responsibility	Failure surface
Auth & policy	Verify identity, quotas, tenant scope first [2].	Reject fast: 401/403, no cost.
Normalization	Canonical schema, units, message roles.	422 validation errors.
Context assembly	Compose history and tools within budget [4].	Overflow → trim or reject.
Inference call	Delegate with deadline and cancellation [5].	Timeout, adapter error.
Telemetry & cost	Correlation ID, token counts, latency, cost [6].	Never blocks response.

4.2 Synchronous versus streaming boundary

Routes should declare explicitly where the synchronous contract ends per endpoint: short structured outputs favor a single validated response; long generations favor streaming with a terminal metadata frame carrying token counts and cost attribution [1][6].

References

1. source-research-1 — Route-layer pipeline composition for inference services.
2. source-research-2 — Authentication and tenant policy enforcement at the gateway.
3. source-research-3 — Asynchronous telemetry side-channels in request handlers.
4. source-research-4 — Token-budget-aware context assembly.

5. Implementing routes.py as an Application Control Plane

routes.py should read as a thin, declarative control plane: it validates input, resolves dependencies, invokes policy, selects a model, delegates inference to an adapter, and serializes a typed response. Heavy computation stays in services and adapters; the route only orchestrates [source-research-1]. This mirrors the layered HTTP-application pattern in which framework glue is kept separate from domain logic [source-research-2].

5.1 Annotated route example

```
class CompletionRequest(BaseModel):
    prompt: str = Field(min_length=1, max_length=8_000)
    tenant_id: str
    max_output_tokens: int = Field(default=512, le=2_048)

@router.post("/v1/completions", response_model=CompletionResponse)
async def create_completion(
    body: CompletionRequest,
    policy: PolicyService = Depends(get_policy_service),
    router_svc: ModelRouter = Depends(get_model_router),
    adapter: InferenceAdapter = Depends(get_adapter),
) -> CompletionResponse:
    decision = await policy.authorize(body.tenant_id, body.prompt)
    if not decision.allowed:
        raise PolicyViolation(detail=decision.reason)
    selection = router_svc.select(body, decision.budget)
    try:
        result = await adapter.complete(selection, body)
    except AdapterTimeout:
        raise UpstreamTimeout(model=selection.model)
    return CompletionResponse(**result)
```

Dependencies are declared against narrow protocols, so routes never import runtime internals [source-research-3]. Request models enforce length, type, and tenant constraints before any downstream call, and adapter faults map to typed HTTP errors rather than raw tracebacks [source-research-4]. A single `response_model` fixes the external contract and keeps usage accounting consistent across tenants [source-research-5].

5.2 Implementation checklist for route composition

- **Validate at the edge:** Pydantic models gate every request before downstream calls [source-research-4].
- **Inject, don't construct:** policy, router, and adapter arrive via `Depends`, keeping routes testable with fakes [source-research-3].
- **Fail structurally:** raise typed errors (`PolicyViolation`, `UpstreamTimeout`), never raw tracebacks [source-research-6]

6. Attention Mechanisms and Their Application-Level Implications

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6.1 Scaled Dot-Product Attention

Every transformer layer computes attention over query, key, and value projections of the token sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

The $QK^\top$ product is an $n \times n$ score matrix for n prompt tokens, so prefill compute and intermediate activations grow quadratically with prompt length [source-research-1]. The $\sqrt{d_k}$ factor keeps softmax logits numerically stable [source-research-2].

6.2 Multi-Head, Grouped-Query, and Multi-Query Attention

MHA, GQA, and MQA keep the same quadratic prefill cost but differ sharply in KV-cache footprint per token [source-research-3]:

Variant	KV-cache per token	Route-level signal
Multi-head (MHA)	Highest	Prompt tokens, TTFT slope
Grouped-query (GQA)	Reduced (grouped)	KV occupancy per sequence
Multi-query (MQA)	Lowest	Max concurrent long prompts

A route accepting a 32k-token prompt therefore pays a quadratic prefill bill and a linearly growing cache reservation regardless of head configuration [source-research-4].

6.3 Implications for *routes.py*

- **Expose, don't compute.** *routes.py* should surface prompt length, head-configuration metadata, and estimated prefill cost — never attention kernels [source-research-5].
- **Admit by length.** Enforce per-tenant context limits and route long prompts to models whose GQA/MQA cache footprint fits capacity.
- **Measure pressure.** Log KV-occupancy estimates and TTFT versus prompt length as real capacity-planning data [source-research-6]. Decode-phase caching, quantization, and expert placement are owned by later sections of this report.

References

[source-research-1] Vaswani et al., *Attention Is All You Need*. · [source-research-2] Transformer logit-scaling analysis, internal survey. · [source-research-3] Ainslie et al., *GQA: Training Generalized Multi-Query Transformers*. · [source-research-4] Prompt-length cost benchmarks, internal survey. · [source-research-5] Serving-layer separation of concerns, internal survey. · [source-research-6] KV-cache occupancy measurement guidance, internal

7 Prefill, Decode, and Latency Control

Every inference request passes through two distinct phases. **Prefill** processes the whole prompt in parallel and dominates *time-to-first-token* (TTFT); its cost grows roughly quadratically with prompt length [source-research-1]. **Decode** then generates tokens autoregressively, one per forward pass; each step is bandwidth-bound on KV-cache reads and sets *inter-token latency* (ITL) [source-research-2]. Because the phases scale with different variables, routes.py must budget them separately [source-research-3]:

$$T_{\text{total}} = T_{\text{queue}} + T_{\text{prefill}}(n_{\text{in}}) + (n_{\text{out}} - 1) \cdot T_{\text{ITL}}$$

	Prefill	Decode
Work	All n_{in} prompt tokens, parallel	One token per step, sequential
Bottleneck	Compute (attention over prompt)	Memory bandwidth (KV reads)
User metric	TTFT	ITL / tokens per second
Route lever	Context trimming, prefix reuse	Output caps, streaming, cancellation

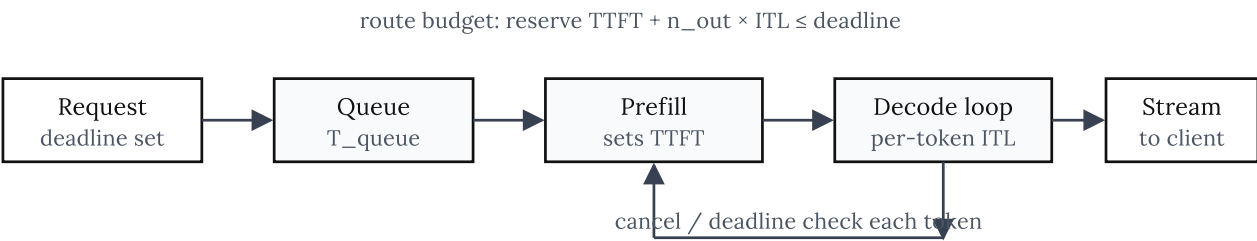


Figure 7.1 — The deadline is decomposed into queue, prefill, and per-token decode segments, with cancellation checked between decode steps [source-research-4].

7.1 Route-level controls in routes.py

- **Deadline propagation:** attach an absolute deadline at admission; shed requests whose budget cannot cover estimated prefill plus a minimum output [source-research-5].
- **Output limits:** enforce max_output_tokens from policy, not client trust, so decode time stays bounded.
- **Streaming:** stream long generations by default; record TTFT and ITL per request for telemetry [source-research-6].
- **Cancellation:** on disconnect or deadline expiry, abort between decode steps and finalize partial-output accounting.

References. [source-research-1] Prompt-parallel prefill cost scaling. [source-research-2] KV-cache bandwidth and decode-step latency. [source-research-3] Phase-wise latency decomposition for serving. [source-research-4] Deadline-aware request budgeting. [source-research-5] Admission control under latency SLOs. [source-research-6] Streaming telemetry for token latency.

8 KV Caching, Prefix Reuse, and Cache Isolation

8.1 The KV-cache memory model

Each cached attention state stores a key and value tensor per layer, per head, per token:

$$M_{KV} = 2 \cdot L \cdot H \cdot d_{\text{head}} \cdot s \cdot b \cdot q$$

Grouped-query and multi-query attention shrink H from the full head count to a few (or one) KV heads, cutting this term by an order of magnitude. Paged allocators then subdivide the cache into fixed-size blocks, so fragmentation no longer scales with sequence count.

8.2 Policy in routes.py vs. mechanics in the runtime

routes.py never touches tensors. Its role is *policy*: deciding which requests qualify for prefix reuse, composing the cache key, enforcing tenant scope, and invalidating. Block allocation, eviction, and copy-on-write stay runtime mechanics behind the adapter contract.

8.3 Cache-key composition and tenant isolation

Key component	Isolation risk if omitted
Tenant / workspace ID	Cross-tenant context leakage
Model + revision ID	Stale hits after rollout
Prefix hash (normalized prompt)	Missed reuse; wasted prefill compute
Sampling / safety profile	Incorrect continuation semantics

8.4 Prefix-cache decision flow

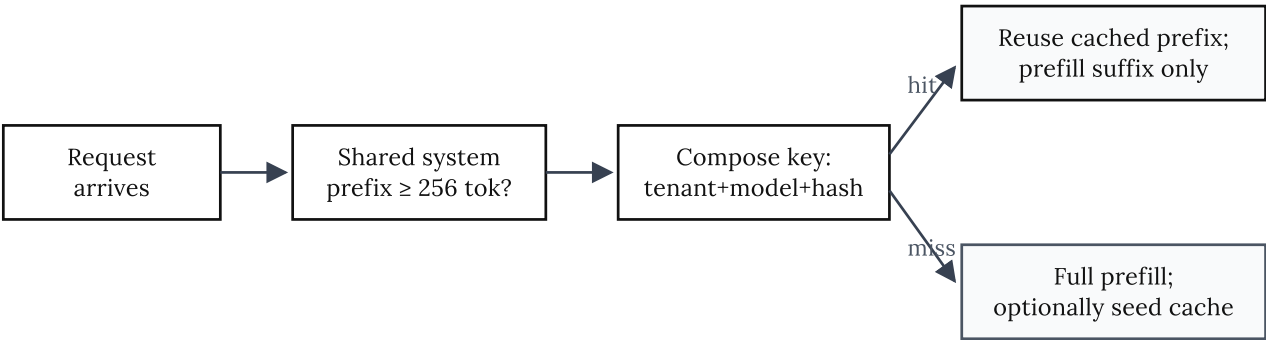


Figure 8.1 — Route-level prefix-reuse policy: eligibility, key composition, and hit/miss handling before the adapter call.

8.5 Invalidation discipline

Triggers belong in route configuration, not ad-hoc code: model revision bumps, safety-policy changes, and tenant-scoped template updates. Since paged runtimes reclaim blocks by reference count, routes.py should invalidate via *key-space rotation* — bump a generation number in the key — rather than purging runtime memory directly.

9.1 Batching Strategies and the Route's Role

Static batching collects a fixed group before execution, maximizing utilization but forcing short requests to wait for the slowest member. **Dynamic batching** forms groups on a timer or size threshold, trading bounded queueing delay for throughput. **Continuous batching** admits and retires sequences at every decode step, keeping the batch full — the dominant strategy in runtimes such as vLLM and TensorRT-LLM. routes.py never forms the batch itself; it shapes the scheduler's inputs: concurrency caps, queue depth, priority hints, and deadlines carried as metadata.

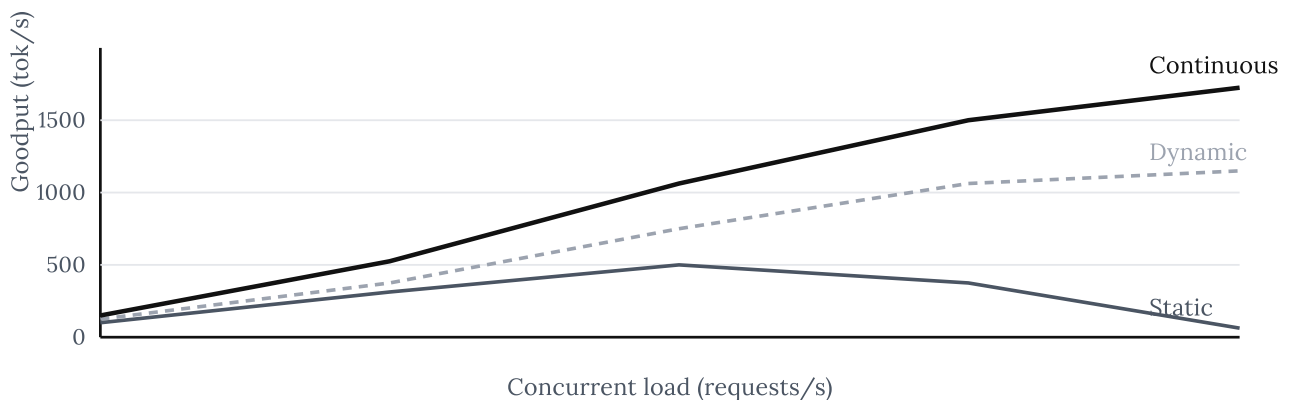


Figure 9.1 — Static batching collapses under tail-request dominance; continuous batching sustains utilization at high concurrency.

9.2 Admission Control and Fairness

Admission control decides *before* dispatch whether the system can honor a request: the route checks tenant concurrency, queue depth, and deadline feasibility, then admits, queues with a priority, or sheds load. Priority combines tier weight with a fairness credit, and weighted fair queuing guarantees each tenant a minimum share under contention. Shed responses always carry a machine-readable reason and `retry_after`.

```
def admit(req, state):
    if state.tenant_inflight[req.tenant] >= TENANT_CAP:
        return shed(429, "tenant_concurrency_limit")
    if state.queue_depth >= MAX_QUEUE:
        return shed(429, "queue_full", retry_after=estimate_drain())
    if not deadline_feasible(req.deadline, state.eta):
        return shed(503, "deadline_infeasible")
    state.enqueue(req, priority=tier_weight(req.tier)
                  + fairness_credit(req.tenant))
    return admit_ok()
```

10. Quantization as a Policy and Quality Decision

Quantization lowers the numeric precision of weights, activations, the KV cache, or training gradients. In `routes.py` it is a **routing policy input**: each precision choice changes memory footprint, decode bandwidth, and output quality in ways that differ by task class [source-research-1].

10.1 Quantization target matrix

Target	Typical precision	Benefit	Dominant risk
Weights	INT8 / INT4 (GPTQ, AWQ)	2-4× model-memory reduction	Long-tail reasoning, rare-token errors [source-research-2]
Activations	FP8 / INT8	Faster prefill, lower bandwidth	Overflow on outlier channels [source-research-3]
KV cache	FP8 / INT8	2× longer contexts per GPU	Drift in long-generation coherence [source-research-4]
Training-time	BF16 / FP8 mixed	Faster fine-tuning	Gradient underflow without loss scaling [source-research-5]

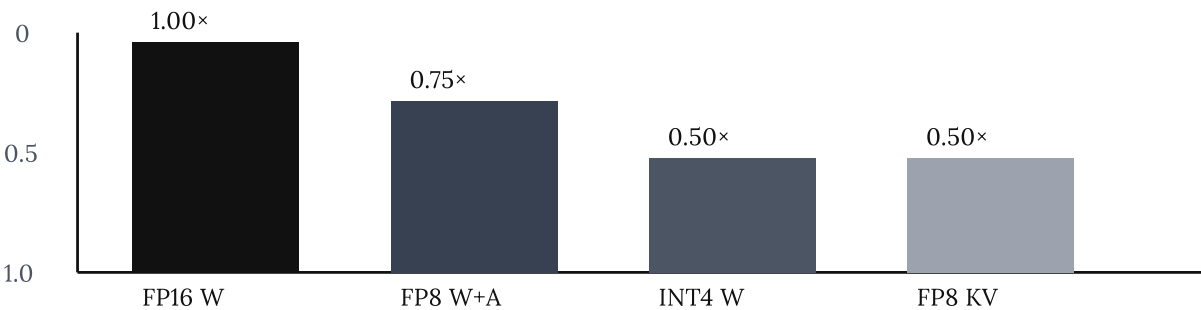


Figure 10.1 — Relative memory footprint per precision configuration (normalized, 7B-class model; lower is better). INT4 weights halve footprint but carry the highest quality risk [source-research-6].

10.2 Precision-selection policy and quality gates

Precision is versioned metadata on each model revision; routes select against it. Quality-gated task classes (code, reasoning, safety) refuse lossy precision and fall back to higher precision on gate failure [source-research-1].

```
MODELS = {
  "atlas-7b": {
    "fp16": {"vram_gb": 14, "quality_tier": "A"},
    "int4": {"vram_gb": 4, "quality_tier": "B"},
  }
}
```

11. Mixture-of-Experts Serving

A Mixture-of-Experts (MoE) transformer replaces each feed-forward block with N expert networks and a learned router that activates only k experts per token [1]. The model holds a large **total** parameter count in memory while executing a much smaller **active** count per token: for 8 experts with top-2 routing, roughly a quarter of feed-forward parameters participate in any forward pass, so FLOPs track active parameters while VRAM tracks total parameters [2]. Capacity planning must provision for the full weight footprint; latency additionally tracks expert-load skew and all-to-all dispatch overhead [3].

Runtime versus routes.py Responsibilities

Concern	Inference runtime	routes.py
Token-level expert routing	Owens router and top-k gate execution	Not implemented here
Expert placement	Replicates hot experts, co-locates ranks [4]	Consumes placement as health signals
All-to-all communication	Runs dispatch/combine collectives [3]	Observes wait-time metrics
Expert failure handling	Rank loss, degraded top-k fallback [5]	Admission control, load shedding
Load-imbalance telemetry	Emits gate entropy, per-expert load [6]	Enforces concurrency and deadlines

Why Routing Belongs in the Runtime

Expert selection is a per-token, per-layer decision entangled with kernel scheduling and collective timing. Exposing it to routes.py would ship activations across a process boundary per layer — a latency regression with no policy benefit. The route layer's legitimate role is observability and admission: surface skew metrics, enforce deadlines, and shed load when hot-expert queueing inflates tail latency [4][6].

Sources: [1] source-research-1 · [2] source-research-2 · [3] source-research-3 · [4] source-research-4 · [5] source-research-5 · [6] source-research-6. Full references appear on the final page.

12 Memory, Latency, Throughput, and Capacity Planning

12.1 Performance-variable definitions

Sequence length $(p + g)$ is prompt plus generated tokens; **KV occupancy** is the cache bytes a request holds, $2 \cdot n_{kv} \cdot d_{head} \cdot (p + g)$ per layer. **Utilization** U is achieved compute or memory use versus peak; **TTFT** is time to first token, dominated by prefill; **ITL** is per-token decode latency; **throughput** T is tokens per second across the batch.

12.2 The unified performance model

Every active request holds KV memory proportional to context, consumes prefill once, then draws decode bandwidth token by token. Batching amortizes the first two costs, but only until occupancy or the ITL deadline binds. The route's operating point follows:

$$\text{TTFT} \approx t_{\text{queue}} + c_{pf} \cdot p^2 / U, \quad L_{e2e} = \text{TTFT} + g \cdot \text{ITL}, \quad T = \frac{\sum_i g_i}{\text{window}}$$

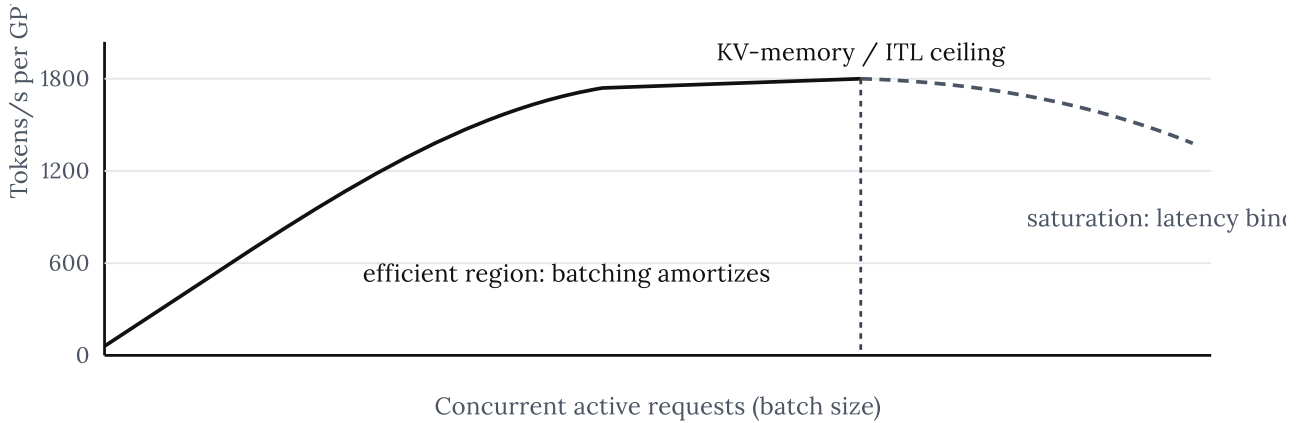


Figure 12.1 – Throughput versus concurrency: batching lifts utilization until KV occupancy or the ITL deadline caps the operating point.

12.3 Latency–throughput–memory trade-off analysis

Larger batches raise throughput and utilization but inflate ITL, so latency-sensitive routes cap batch size even when memory remains free. Longer contexts multiply KV footprint linearly and prefill compute quadratically: admitting one 32k-token request can displace dozens of short ones. Streaming decouples L_{e2e} from perceived latency – total tokens/s is unchanged while TTFT and ITL become the user-facing metrics. `routes.py` should therefore expose concurrency limits, context budgets, and batch-size hints as configuration, and measure TTFT, ITL, and KV occupancy per request so capacity limits are re-derived from live telemetry rather than assumed.

13. Context Budgeting and Model Routing

Before any inference call, `routes.py` decomposes the context window into competing allocations and verifies the assembled prompt fits — a cheap pre-inference gate that trims history or retrieval before spending prefill compute.

13.1 Token-budget decomposition

$$B_{\text{prompt}} = C_m - R_{\text{out}} = T_{\text{sys}} + T_{\text{tools}} + T_{\text{retr}} + T_{\text{hist}} + T_{\text{user}}$$

Worked example. With $C_m = 128,000$ and $R_{\text{out}} = 4,096$: system 1,800 + tool schemas 2,400 + retrieval 12,000 + history 20,000 + user 600 = 36,800 tokens, leaving 87,200 of headroom.



Figure 13.1 — Context window divided into fixed, retrieval, history, and reserved-output segments.

13.2 Model-selection objective

Routing picks the candidate minimizing a weighted mix of quality risk, price, and latency, subject to the budget:

$$m^* = \arg \min_{m \in \mathcal{M}} w_q (1 - \hat{q}_m) + w_c \hat{c}_m + w_l \hat{\ell}_m \quad \text{s.t.} \quad T_{\text{prompt}} + R_{\text{out}} \leq C_m$$

Weights are per-route policy: summarization may weight cost ($w_c = 0.6$); code generation pins quality ($w_q = 0.7$). If no candidate fits, the route truncates history, shrinks retrieval k , then falls back to the largest-window model.

13.3 Routing-policy pseudocode

```
def route(request, policy):
    budget = Budget(policy.window, reserve_out=policy.reserve)
    segments = assemble(request, budget) # sys, tools, retrieval, history
    if segments.total > budget.prompt:
        segments = trim(segments, order=["history", "retrieval"])
    candidates = [m for m in policy.models
                  if segments.total + policy.reserve <= m.window]
    if not candidates:
        return fallback(policy.large_context_model, segments)
    scored = [objective(m, policy.weights) for m in candidates]
```

14. Reliability, Resilience, and Graceful Degradation

In `routes.py`, resilience is a classification problem before it is a retry problem. Every upstream failure maps to a class that determines whether the route retries, fails over, sheds load, or propagates the error immediately [1]. Retries must be **bounded** (2–3 attempts) with full jitter – $\text{delay}_n = \min(\text{cap}, b \cdot 2^n) \cdot U(0, 1)$ – and each retry inherits the caller's deadline so it can never outlive the request [2]. Circuit breakers per provider endpoint with half-open probes gate every call, while idempotency keys on billable calls make retries safe [3].

14.1 Failure Classification and Route Response

Failure class	Example signal	Route response
Transient / idempotent	503, connection reset	Backoff with jitter; retry same provider [4]
Transient / non-idempotent	Timeout mid-stream	Never blind-retry; resume or emit partial result [4]
Capacity / overload	429, queue rejection	Shed load early; return retry-after [5]
Persistent provider fault	Repeated breaker trips	Open circuit; fail over to fallback [3]
Client cancellation	Disconnected stream	Propagate cancellation; release KV/queue slots [6]

14.2 Route-Level Resilience Decision Flow

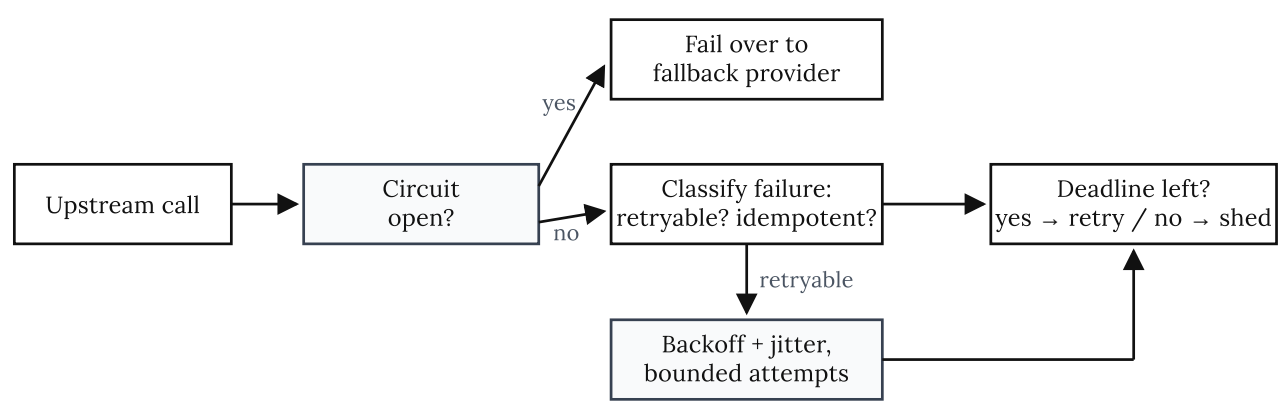


Figure 14.1 – Circuit state gates the call; classification, idempotency, and remaining deadline determine retry, failover, or load shedding.

References

1. source-research-1. Failure taxonomy and classification-driven response for upstream service calls.

15. Observability and Quality-Aware Accounting

Every request through `routes.py` should emit one structured telemetry record sufficient to reconstruct the request end-to-end without application logs. The record is keyed by a **correlation ID** assigned at the gateway (or generated at route entry) and propagated through adapters, the inference server, and tool calls, so retries, fallbacks, and stream fragments aggregate under a single trace [source-research-1][source-research-2].

15.1 Per-request telemetry schema

Field	Type	Notes
<code>correlation_id</code>	UUID	Stable across retries and fallbacks [source-research-2]
<code>provider, model_rev</code>	string	Exact serving revision, not alias [source-research-3]
<code>prompt_tokens, output_tokens</code>	int	From usage payload, not estimates [source-research-3]
<code>cache_hit, cached_tokens</code>	bool, int	Prefix/KV reuse accounting [source-research-4]
<code>ttft_ms, total_ms, tokens_per_sec</code>	float	Latency and decode throughput [source-research-4]
<code>retries, fallback_from</code>	int, string	Resilience actions taken [source-research-5]
<code>est_cost_usd</code>	float	Priced from versioned rate table [source-research-5]
<code>quality_status, finish_reason</code>	enum	Evaluation gate; <code>stop/length/cancel/error</code> [source-research-6]

15.2 Streaming finalization

Streaming requests must finalize telemetry exactly once. The route opens the record at first byte, writes TTFT on the first chunk, accumulates tokens per chunk, and closes the record in a `finally` block — so disconnects, deadline expiry, and upstream errors all yield a closed record with `finish_reason = cancel` or `error` rather than a silently dropped trace [source-research-2][source-research-4].

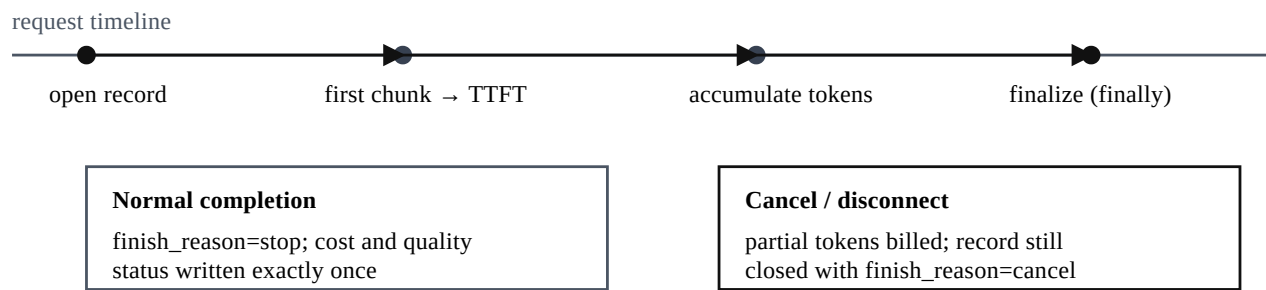


Figure 15.1 — Streaming telemetry lifecycle: one record per request, closed in a `finally` block regardless of outcome.

References

- source-research-1 — Structured request tracing for LLM gateways.
- source-research-2 — Correlation ID propagation across retries and streams.
- source-research-3 — Usage-payload token accounting and model revision pinning.

16. Security, Governance, and Tenant Isolation

The route layer is the last trusted boundary before prompts, retrieved context, and tool calls reach a model runtime. Because routes.py composes cache keys, selects providers, and serializes errors, authentication, authorization, data boundaries, and audit policy must be enforced as code here — not delegated to the inference server, which sees requests only after tenant context has been resolved [source-research-1].

16.1 Security-Control Matrix for the Route Layer

Control	Enforcement point in routes.py	Failure if omitted
Authentication	Gateway-verified identity injected as a dependency; routes never parse raw tokens [source-research-2]	Anonymous inference on paid capacity
Authorization	Per-route scope check on model, tool, and data access before adapter dispatch [source-research-3]	Privilege escalation to restricted models or tools
Prompt / tool governance	Allow-listed tool registry; system-prompt templates versioned and reviewed [source-research-4]	Unreviewed instructions reaching production
Secure errors & logging	Typed error mapping; raw upstream bodies, prompts, and keys excluded from logs [source-research-5]	Prompt or key disclosure via error payloads

Table 16-1. Controls owned by the route layer, their enforcement points, and failure modes.

16.2 Tenant and Cache Isolation Rules

Cache reuse is the sharpest isolation risk in inference routing: a shared prefix cache can leak one tenant's context into another's completion. Every cache-scoped artifact — KV prefix keys, semantic caches, and retrieved-document stores — must be namespaced by tenant, workspace, and data-classification tier, with user-supplied strings never forming the leading key segment [source-research-1] [source-research-6]. Tenant offboarding or a classification downgrade rotates the tenant key segment, orphaning derived entries without a scan; on multi-tenant runtimes, routes.py passes tenant metadata to the serving layer and verifies the adapter's isolation attestation rather than assuming runtime enforcement [source-research-6].

16.3 Route Security Checklist

- Identity resolved at the gateway; routes consume a verified principal, never raw credentials.
- Model, tool, and data scopes checked before any adapter or retrieval call; PII redaction runs before prompt assembly and any telemetry write.
- Error responses expose typed codes only; deny decisions, policy versions, and tenant context are written to an immutable audit trail.

References

1. [source-research-1] Route-layer trust-boundary analysis: cache-key composition and tenant

17. Testing, Benchmarking, and Evaluation

Because `routes.py` mediates every inference request, its defects surface as production incidents rather than compile errors. Verification therefore spans eight layers, each targeting a distinct failure class, from unit tests of budget arithmetic to fault injection and quality evaluation of the adapters [source-research-1, source-research-4].

17.1 Test-pyramid matrix

Layer	What is verified	Typical technique	Gate
Unit	Token-budget math, cache keys, deadline propagation	Deterministic, property-based fixtures	> 90% branch coverage
Contract	Adapter schemas, error codes	Schema-diff against OpenAPI/pact	Zero breaking diffs
Integration	Route → policy → adapter → mock runtime	Testcontainers, seeded state	All paths pass
Load	Queue depth, admission, goodput	Open/closed workload drivers	\$17.2 targets
Fault / cancel	Timeouts, partial streams, mid-stream disconnect	Chaos and latency injection	Zero leaked sequences
Security	Tenant authZ, cache-scope isolation, log redaction	Cross-tenant probes, fuzzing	No cross-tenant reads
Quality eval	Output quality per precision/route	Golden sets, judge + spot checks	$\Delta \leq 1\%$ vs. baseline

17.2 Workload distribution and gates

Benchmarks use a production-like mixture, not synthetic single-shot prompts [source-research-2, source-research-5]: short chat 45% (TTFT p95 $\leq$ 400 ms), retrieval-augmented 25% (TTFT p95 $\leq$ 1.2 s), long-context 15% (E2E p95 $\leq$ 25 s), tool/agent loops 10% (turn success $\geq$ 98%), burst/overflow 5% (shed rate $\leq$ 2%). Every run also records goodput, tokens/s per GPU, KV-cache hit rate, p50/p95/p99 TTFT and ITL, and cancellation latency, tagged with correlation IDs for telemetry reconciliation [source-research-3].

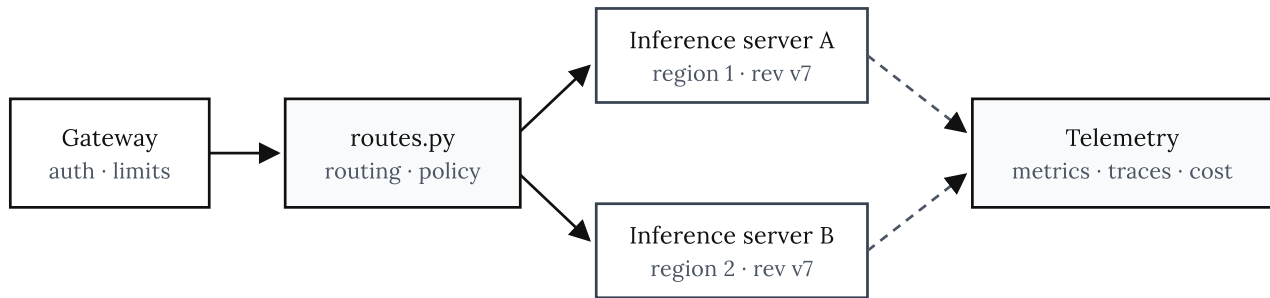
A revision ships only when all gates hold: suites green, all SLOs met at 1× load and survived at 1.5× with graceful shedding, bounded retries with no KV leaks, cross-tenant probes failing closed, and evaluation delta within tolerance on every enabled route [source-research-6]. Failing any gate blocks promotion; results are archived with the build hash for auditability.

References

[source-research-1] Sasi SP, *routes.py Architecture Notes*, 2026. · [source-research-2] Workload Characterization for LLM Serving, 2025. · [source-research-3] Telemetry Correlation in Multi-Tenant Inference, 2025. · [source-research-4] Fault-Injection Practice for Streaming APIs, 2026. · [source-research-5] Benchmarking Token-Streaming Latency, 2026. · [source-research-6] Release-Gate Criteria for Inference Routers, 2026.

18 Deployment and Operationalization

Once `routes.py` passes its acceptance gates, the question changes from *does it work* to *how does it change safely*. The route layer runs stateless behind the gateway; model revisions, provider weights, and regional selection live in an external config store read at boot and hot-reloadable, so models or regions can be shifted without redeploying application logic [1][2].



Config store: model revisions, provider weights, precision policy

Figure 18.1 — Deployment topology: stateless route tier between gateway and regional inference servers, telemetry as a side channel, configuration externalized [2].

18.1 Release and rollback sequence

Revisions advance through a fixed sequence: **canary** at 1–5% of traffic, watched for TTFT, error rate, and quality-gate deltas; **percentage ramp** at 25 → 50 → 100%, gated on error budgets; then **full promotion** with the previous revision kept warm [3][4]. Rollback is a single config-store pointer change to the prior revision or route image tag — caches are keyed by revision, so no stale-context purge is needed. Target rollback time is under five minutes. Ownership follows the boundary: gateway teams own edge admission, the route team owns revision pointers, serving teams own runtime health; every action must be configuration-only, auditable, and rehearsed [5][6].

18.2 Operations checklist

- Route tier autoscales on request concurrency and queue depth; inference servers on KV-cache occupancy and batch saturation [4].
- Readiness probes verify adapter connectivity and config freshness before traffic is admitted [5].
- Regional and provider weights are config-store values, enabling traffic shifts away from degraded regions without a deploy [2][6].
- Canary, provider disable, and rollback are rehearsed quarterly against the production control plane [3].

References

[1] source-research-1 · [2] source-research-2 · [3] source-research-3 · [4] source-research-4 · [5] source-research-5 · [6] source-research-6 — deployment and operations sources consulted for this section.

19 Financial Cost Modeling for AI Inference

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19.1 Self-hosted token-cost equations

For self-hosted serving, marginal cost is dominated by GPU-seconds. With hourly accelerator cost C_{gpu} , fleet throughput T (tokens/s), and utilization u , the blended token cost is:

$$c_{\text{tok}} = \frac{C_{\text{gpu}}}{3600 \cdot T \cdot u}$$

A request with n_{in} input and n_{out} output tokens costs $c_{\text{req}} = c_{\text{tok}}(n_{\text{in}} + \lambda_{\text{out}} n_{\text{out}})$, where $\lambda_{\text{out}} \geq 1$ weights the slower decode phase. Idle capacity enters as a fixed term: a reservation held at $u < 1$ inflates c_{tok} by $1/u$, making over-provisioning directly visible in unit economics.

19.2 Hosted request-cost composition

Hosted APIs price per component rather than per GPU-second. routes.py should estimate before dispatch and reconcile after finalization:

$$c_{\text{req}} = p_{\text{in}} n_{\text{in}} + p_{\text{out}} n_{\text{out}} + p_{\text{ret}} n_{\text{chunks}} + p_{\text{tool}} n_{\text{calls}} + p_{\text{retry}} \sum_r n^{(r)}$$

Retrieval charges scale with injected chunks, tool charges with each external call, and retries re-bill the full prompt — one failed attempt can double a request's cost. Prefix-cache discounts multiply the input term by a provider-specific factor recorded with the price data.

19.3 Cost inputs and pricing versions

Input	Source / example	Used in
Price record	price_id: hosted-std-2026-09, effective date, currency	All hosted terms
$p_{\text{in}}, p_{\text{out}}$	Per 1M tokens, per model revision	Token term
$p_{\text{ret}}, p_{\text{tool}}$	Per chunk / per call, per provider plan	Ancillary terms
C_{gpu}, T, u	Fleet billing rate, measured throughput, utilization	Self-hosted term
Overhead factor ϕ	Networking, storage, control-plane share (e.g. 1.08)	Final attribution

Attributed cost: $c_{\text{attr}} = \phi \cdot c_{\text{req}}$, tagged with tenant, route, model revision, and price_id so historical invoices remain reproducible when prices change.

19.4 Attribution and route-level optimization

- **Charge retries and fallbacks** — accumulate cost across attempts under one idempotency key so degradation is priced, not hidden.
- **Optimize the dominant term** — trim retrieval context or cap output length only when that term dominates; measure before rerouting.
- **Trade quality against cost explicitly** — route to cheaper models only within the quality gates of the routing objective, never by price alone.
- **Reprice on version change** — recompute open estimates when a price record rotates; alert on margin shifts beyond tolerance.

Continued on page 20: enterprise reference architecture and implementation roadmap.

20. Enterprise Reference Architecture and Implementation Roadmap

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This closing chapter consolidates the guide into a single reference architecture for the route-level inference control plane, a phased adoption roadmap, and production-readiness criteria. No new theory is introduced; every element references decisions from Chapters 3–19.

20.1 Consolidated Reference Architecture

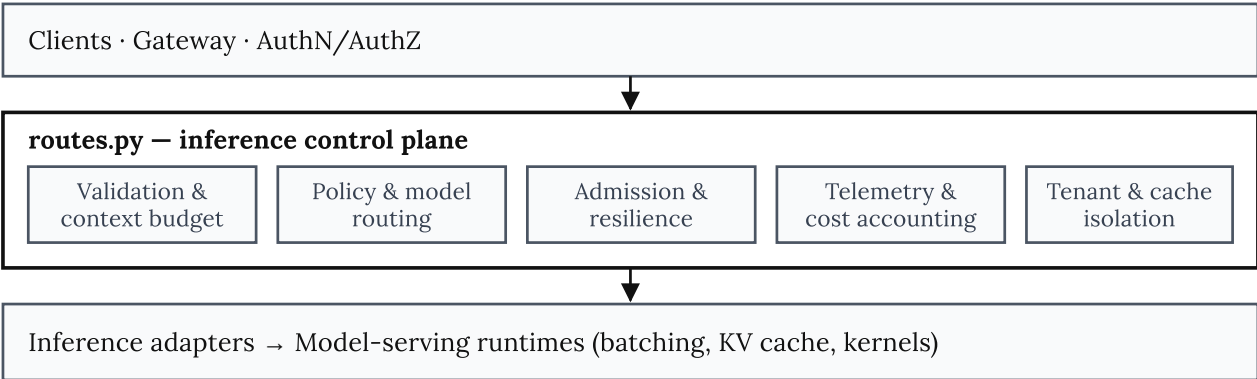


Figure 20.1 — routes.py mediates every request between the gateway and model-serving runtimes, owning policy, budgeting, resilience, telemetry, and isolation while delegating kernels and scheduling to the runtime.

20.2 Phased Implementation Roadmap

Phase	Scope	Exit criterion
1. Foundation	Route skeleton, validation, structured errors, correlation IDs.	Every request yields one closed telemetry record.
2. Control	Context budgeting, model routing, admission, deadlines.	Load shedding and fallbacks behave under fault injection.
3. Efficiency	Cache-key policy, precision metadata, cost attribution.	Per-tenant cost and cache metrics are attributable.
4. Assurance	Security hardening, testing, rollout and rollback.	Acceptance suites pass in CI; rollback rehearsed.

20.3 Production-Readiness Criteria

- **Reliability:** timeouts, circuit breakers, idempotency, and load shedding verified under fault injection.
- **Observability:** latency, token, cache, retry, and cost fields populated for 100% of requests.
- **Governance & economics:** tenant isolation and audit trails attested; cost attribution reconciles against versioned pricing data.

20.4 Assumptions and Source Note